RELATED OBJECTS:

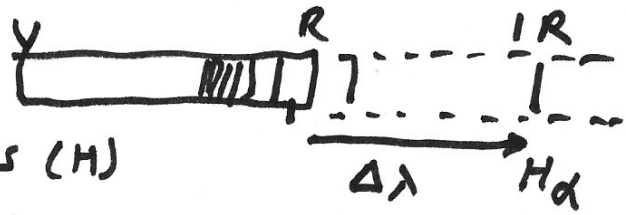
QUASARS

1950s: RADIO ASTRONOMY BEGINS

3RD CAMBRIDGE CATALOG OF RADIO SOURCES

3C 273 VISIBLE LIGHT: ← "DIM STAR"

SPECTRUM:



→ BALMER SERIES (H)

HUGE REDSHIFT! MOVING AWAY FROM US.

DOPPLER FORMULA: $v = c\left(\frac{\Delta\lambda}{\lambda}\right)$ $v = 15\% c$!

HUBBLE LAW: $v = H_0 d$ $d = 2$ BILLION ly!

3C 273 IS AS BRIGHT AS ABOUT 100 MILKY WAY GALAXIES!

QUASI-STELLAR OBJECT

→ QUASAR!

TODAY: 1,000s OF QUASARS ARE KNOWN

ALL ARE VISUALLY BRIGHT (LIKE 3C 273)

ALL ARE XRAY BRIGHT (ENERGETIC)

ONLY 10% ARE RADIO BRIGHT (??)

ALL ARE ABOUT THE SIZE OF OUR SOLAR SYSTEM

WE SEE QUASARS UP TO 13 BILLION ly AWAY.

MOST DISTANT OBJECTS SEEN → AMONG THE FIRST OBJECTS TO FORM IN THE UNIVERSE.

GENERAL RELATIVITY: DOUBLE IMAGES, EINSTEIN CROSSES, EINSTEIN RINGS

· ACTIVE GALAXIES (A.K.A. AGNS)

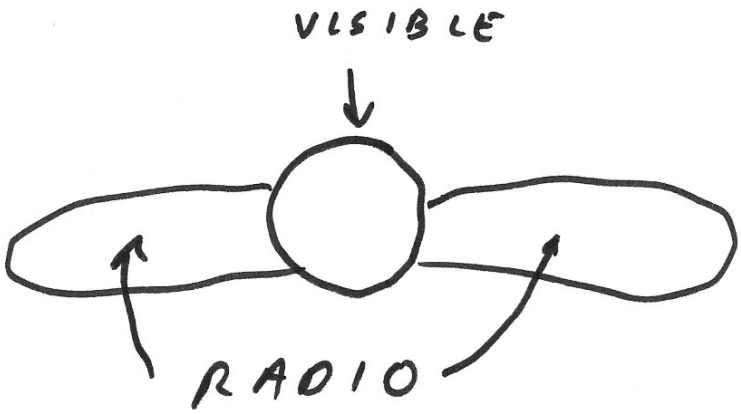
AGNs → ACTIVE GALACTIC NUCLEI

TYPES:

- SEYFERTS : SPIRALS
- BL LACs AKA BLAZARS : ELLIPTICALS
- (DOUBLE LOBED) RADIO GALAXIES

  TYPICALLY GIANT ELLIPTICALS

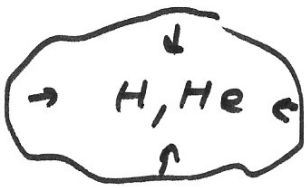
  IN VISIBLE LIGHT THESE APPEAR NORMAL



AGNs TEND TO BE AT "INTERMEDIATE" DISTANCES / AGES IN THE UNIVERSE

GALAXY EVOLUTION, PERHAPS!

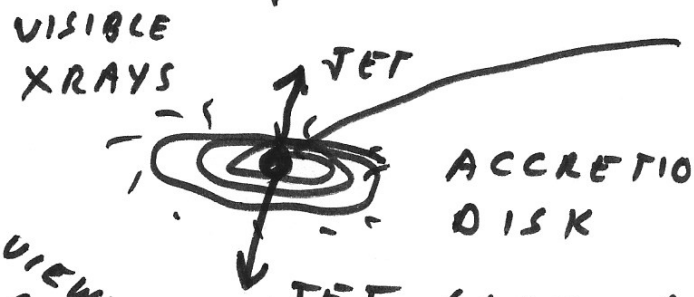
WITHIN THE FIRST BILLION YEARS OF THE UNIVERSE (TIME AFTER THE BIG BANG), THERE WERE MANY VAST CLOUDS OF H, He THAT ARE COLLAPSING:



"DARK AGES"
→ JAMES WEBB TELESCOPE (IR)

↓ TIME

SEE:
QUASAR



↓ TIME

SEE:
ACTIVE GALAXY

STARS FORM
CENTRAL ACTIVITY STARTS TO DECLINE

↓ TIME

SEE:
NORMAL GALAXY

CENTRAL ACTIVITY DECLINES MORE

LONG AGO ⟶ TODAY

TIME: QUASARS → AGNs → NORMAL GALAXIES

WHY SPIRALS OR ELLIPTICALS (?)

OLDER IDEA: ANGULAR MOMENTUM

ANCIENT CLOUDS:

LITTLE A.M.: ELLIPTICAL

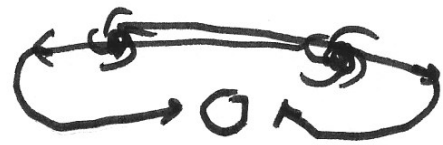
LOTS OF A.M.: SPIRAL

WHY THE DIFFERENCE IN A.M.?

PROBLEM: COMPUTER MODELS SAY ALL SHOULD BE SPIRALS (?)

NEWER IDEA: GALAXY COLLISIONS (?)

2 SPIRALS:



MERGED (ELLIPTICAL)

EVIDENCE? VIRGO CLUSTER

STAR SIZES 2 END, HUBBLE GALAXIES, 1 AY 22 Q/AGNS, ATLAS OF UNIVERSE

COSMOLOGY

THE SCIENCE INVOLVING THE ORIGIN, CONTENTS, STRUCTURE, EVOLUTION AND FATE OF THE ENTIRE UNIVERSE.